

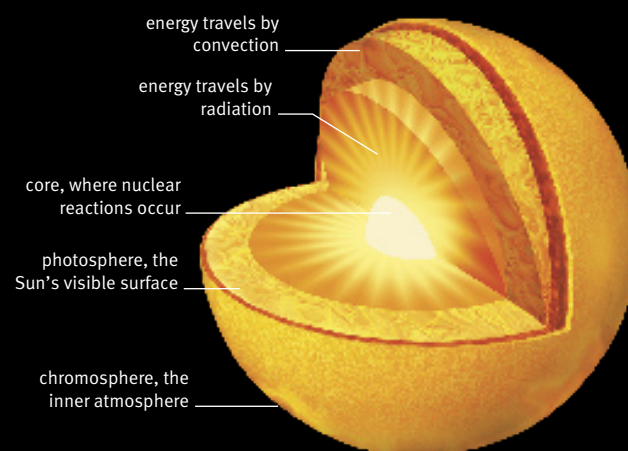
The Sun

The Sun is the closest star to Earth. Like all other stars, it is a vast ball of incredibly hot, brilliant gas. Gravity keeps the Sun together by pulling the gas in toward the center, where it converts hydrogen to helium and in the process produces heat and light. The Sun has been shining for about 4.6 billion years and will do so for about another 5 billion.

FEATURES

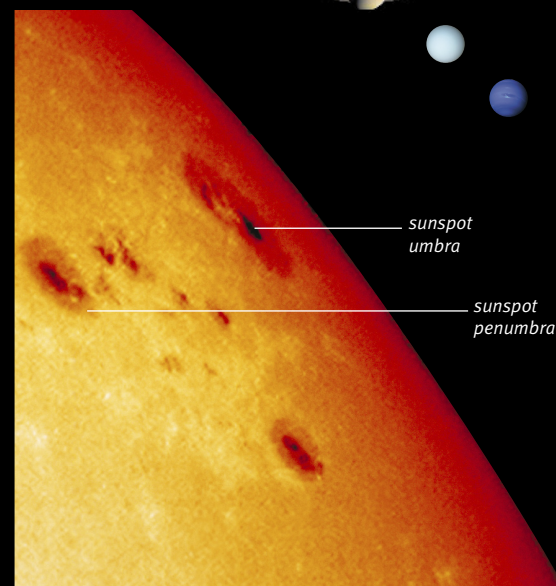
The Sun is about three-quarters hydrogen and a quarter helium, with small amounts of 90 or so other elements. About 60 percent of all this is in the core, where the temperature and pressure are extremely high and nuclear reactions occur.

The Sun is not solid but has a visible surface, the photosphere. The temperature of the photosphere gives the Sun its color. It consists of constantly renewing granules of rising gas, each of which is about 600 miles (1,000 km) across. Beyond the photosphere is the Sun's atmosphere, which is not normally visible. Nearest the Sun is the chromosphere, which extends out about 3,100 miles (5,000 km), and outside of this is the corona, which extends for millions of miles into space.



Inside the Sun

The Sun converts hydrogen to helium at about 600 million tons per second. Energy produced moves through the Sun by radiation, then nearer the surface by convection, and is released via the photosphere.



Sunspots

Dark patches on the Sun's surface are called sunspots. These are relatively cool regions of the photosphere—about 2,700°F (1,500°C) cooler than the surrounding surface. Sunspots appear periodically, usually in pairs or groups, between 40 degrees north and 40 degrees south of the Sun's equator. They are hundreds of thousands of miles wide and last for a few weeks.

OBSERVING THE SUN

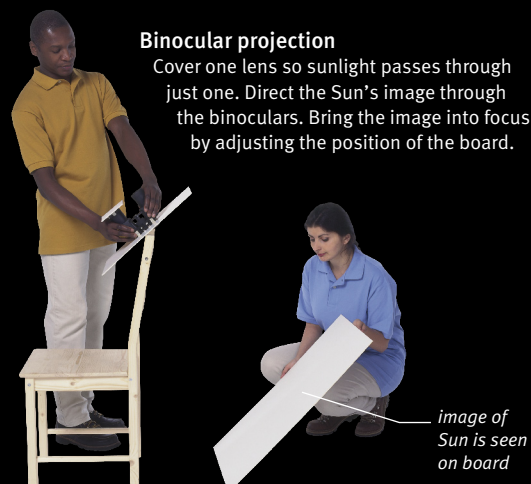
The Sun cannot be observed directly, but it is possible to view its disk safely. One way to do this is by projecting the Sun's image onto white posterboard using binoculars or a telescope.

A total eclipse of the Sun is a dramatic event that offers the opportunity to see the Sun's outer atmosphere and prominences flaring from the surface. Spectacular light displays called aurorae (singularly aurora) can be seen

WARNING

Never look at the Sun directly with the naked eye or with any instrument. The light will burn your retina, causing permanent blindness.

by observers in the most northerly and southerly latitudes. These are produced by the interaction between particles from the Sun and Earth's upper atmosphere.



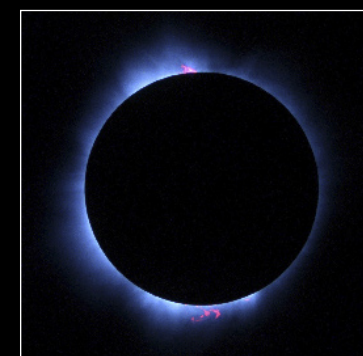
Telescope projection

Position the card about 20 in (50 cm) from the telescope's eyepiece, and aim the telescope at the Sun. To sharpen the Sun's image, adjust the eyepiece. Any sunspots will be seen as blackened dots on the Sun's disk.



Aurora Borealis

Also known as the Northern Lights, the Aurora Borealis can be seen from locations north of about latitude 50°N. The colorful display of glowing gas shown above was captured just after dusk from the Yukon Territory, Canada. The Aurora Australis (Southern Lights) can be seen south of about 50°S.



Total eclipse

The Sun and Moon appear the same size in the sky, so when the Moon passes directly in front of the Sun, the latter's disk is completely hidden. The Sun's outermost layer, its corona, is then visible. In this photo, red prominences are also seen.